

14. The marks obtained by students in an examination follow a normal distribution. If 30% of the students got marks below 35 and 10% got marks above 60, find the mean and percentage of students who got marks between 40 and 50.

[Ans.: 42.23, 13.88, 28%]

15. Fit a normal distribution to the following data:

Class	60-65	65-70	70-75	75-80	80-85	85-90	90-95	95-100
Frequency	3	21	150	335	326	135	26	4

[Ans.: Expected frequency: 3, 31, 148, 322, 319, 144, 30, 3]

5.5 EXPONENTIAL DISTRIBUTION

A continuous random variable X is said to follow exponential distribution if its probability function is given by

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$= 0, \quad x \leq 0$$

where $\lambda > 0$ is called the rate of the distribution.

5.5.1 Memoryless Property of the Exponential Distribution

The exponential distribution has the memoryless (forgetfulness) property. This property indicates that the distribution is independent of its part, that means future happening of an event has no relation to whether or not this even has happened in the past. This property is as follows:

If X is exponentially distributed, and s, t are two positive real numbers then

$$P[(X > s+t)/(X > s)] = P(X > t)$$

Proof:
$$P[(X > s+t)/(X > s)] = \frac{P[(X > s+t) \cap (X > s)]}{P(X > s)}$$

[using conditional probability]

$$= \frac{P(X > s+t)}{P(X > s)}$$

$$= \frac{\int_{s+t}^{\infty} \lambda e^{-\lambda x} dx}{\int_s^{\infty} \lambda e^{-\lambda x} dx}$$

$$\begin{aligned}
 &= \frac{\lambda \left| \frac{e^{-\lambda x}}{-\lambda} \right|_{s+t}^{\infty}}{\lambda \left| \frac{e^{-\lambda x}}{-\lambda} \right|_s^{\infty}} \\
 &= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} \\
 &= e^{-\lambda t} \quad \dots(5.5)
 \end{aligned}$$

$$\begin{aligned}
 P(X > t) &= \int_t^{\infty} \lambda e^{-\lambda x} dx \\
 &= \lambda \left| \frac{e^{-\lambda x}}{-\lambda} \right|_t^{\infty} \\
 &= e^{-\lambda t} \quad \dots(5.6)
 \end{aligned}$$

From Eq. (5.5) and Eq. (5.6),

$$P[(X > s+t)/(X > s)] = P(X > t), \quad \text{for } s, t > 0$$

5.5.2 Constants of the Exponential Distribution

1. Mean of the Exponential Distribution

$$\begin{aligned}
 E(X) &= \int_{-\infty}^{\infty} x f(x) dx \\
 &= \int_0^{\infty} x \lambda e^{-\lambda x} dx \\
 &= \lambda \left| x \cdot \frac{e^{-\lambda x}}{-\lambda} - 1 \cdot \frac{e^{-\lambda x}}{\lambda^2} \right|_0^{\infty} \\
 &= \lambda \cdot \frac{1}{\lambda^2} \\
 &= \frac{1}{\lambda}
 \end{aligned}$$

2. Variance of the Exponential Distribution

$$\text{Var}(X) = E(X^2) - [E(X)]^2 \quad \dots(5.7)$$

$$\begin{aligned}
 E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx \\
 &= \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx
 \end{aligned}$$

$$\begin{aligned}
 &= \lambda \left| x^2 \frac{e^{-\lambda x}}{-\lambda} - 2x \frac{e^{-\lambda x}}{\lambda^2} + 2 \frac{e^{-\lambda x}}{-\lambda^3} \right|_0^\infty \\
 &= \lambda \left(\frac{2}{\lambda^3} \right) \\
 &= \frac{2}{\lambda^2}
 \end{aligned}$$

Substituting in Eq (5.7),

$$\text{Var}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2} \quad \left[\because \mu = \frac{1}{\lambda} \right]$$

3. Standard Deviation of the Exponential Distribution

$$\text{SD} = \sqrt{\text{Var}(X)} = \sqrt{\frac{1}{\lambda^2}} = \frac{1}{\lambda}$$

4. Mode of the Exponential Distribution

Mode is the value of x for which $f(x)$ is maximum.

$$\begin{aligned}
 f(x) &= \lambda e^{-\lambda x}, \quad x > 0 \\
 &= 0, \quad x \leq 0
 \end{aligned}$$

$f(x)$ will be maximum when $e^{-\lambda x}$ is maximum.

Maximum value of $e^{-\lambda x} = 1$, which is at $x = 0$.

Hence, $x = 0$ is the mode of the exponential distribution

5. Median of the Exponential Distribution

If M is the median of the exponential distribution,

$$\int_{-\infty}^M f(x) dx = \frac{1}{2}$$

$$\int_0^M \lambda e^{-\lambda x} dx = \frac{1}{2}$$

$$\lambda \left| \frac{e^{-\lambda x}}{-\lambda} \right|_0^M = \frac{1}{2}$$

$$-(e^{-\lambda M} - 1) = \frac{1}{2}$$

$$-e^{-\lambda M} = \frac{1}{2} - 1 = -\frac{1}{2}$$

$$e^{-\lambda M} = \frac{1}{2}$$

$$-\lambda M \log e = \log \frac{1}{2} = -\log 2$$

$$\lambda M = \log 2$$

$$M = \frac{1}{\lambda} \log 2$$

Example 1

Let X be a random variable with pdf

$$f(x) = \begin{cases} \frac{1}{5} e^{-\frac{x}{5}} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find (i) $P(X > 5)$ (ii) $P(3 \leq X \leq 6)$ (iii) mean (iv) variance.

Solution

$$\lambda = \frac{1}{5}$$

$$(i) P(X > 5) = \int_5^{\infty} f(x) dx$$

$$= \int_5^{\infty} \frac{1}{5} e^{-\frac{x}{5}} dx$$

$$= \frac{1}{5} \left[e^{-\frac{x}{5}} \right]_5^{\infty}$$

$$= - \left[e^{-\frac{x}{5}} \right]_5^{\infty}$$

$$= -(e^{-\infty} - e^{-1})$$

$$= e^{-1}$$

$$= 0.3679$$

$$\begin{aligned}
 \text{(ii) } P(3 \leq X \leq 6) &= \int_3^6 f(x) dx \\
 &= \int_3^6 \frac{1}{5} e^{-\frac{x}{5}} dx \\
 &= \frac{1}{5} \left[e^{-\frac{x}{5}} \right]_3^6 \\
 &= \frac{1}{5} \left[\frac{1}{e^{\frac{6}{5}}} - \frac{1}{e^{\frac{3}{5}}} \right] \\
 &= -\left(e^{-\frac{6}{5}} - e^{-\frac{3}{5}} \right) \\
 &= e^{-\frac{3}{5}} - e^{-\frac{6}{5}} \\
 &= 0.2476
 \end{aligned}$$

$$\text{(iii) Mean } \mu = \frac{1}{\lambda} = \frac{1}{\left(\frac{1}{5}\right)} = 5$$

$$\text{(iv) Variance} = \text{Var}(X) = \frac{1}{\lambda^2} = \frac{1}{\left(\frac{1}{5}\right)^2} = 25$$

Example 2

A random variable has pdf $f(x) = ce^{-2x}$ for $x > 0$. Find (i) $P(X > 2)$

$$\text{(ii) } P\left(X < \frac{1}{c}\right).$$

Solution

Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_0^{\infty} ce^{-2x} dx = 1$$

$$\left[\frac{ce^{-2x}}{-2} \right]_0^{\infty} = 1$$

$$-\frac{c}{2} \left| e^{-2x} \right|_0^{\infty} = 1$$

$$-\frac{c}{2} (e^{-\infty} - e^0) = 1$$

$$\frac{c}{2} = 1$$

$$c = 2$$

$$\therefore f(x) = 2e^{-2x}, \quad x > 0$$

$$(i) \quad P(X > 2) = \int_2^{\infty} f(x) dx$$

$$= \int_2^{\infty} 2e^{-2x} dx$$

$$= 2 \left| \frac{e^{-2x}}{-2} \right|_2^{\infty}$$

$$= - \left| e^{-2x} \right|_2^{\infty}$$

$$= -(e^{-\infty} - e^{-4})$$

$$= e^{-4}$$

$$= 0.0183$$

$$(ii) \quad P\left(X < \frac{1}{c}\right) = P\left(X < \frac{1}{2}\right)$$

$$= \int_0^{\frac{1}{2}} f(x) dx$$

$$= \int_0^{\frac{1}{2}} 2e^{-2x} dx$$

$$= 2 \left| \frac{e^{-2x}}{-2} \right|_0^{\frac{1}{2}}$$

$$= - \left| e^{-2x} \right|_0^{\frac{1}{2}}$$

$$= -(e^{-1} - e^0)$$

$$= -e^{-1} + 1$$

$$= 0.6321$$

Example 3

If X is random variable which follows an exponential distribution with parameter λ with $P(X \leq 1) = P(X > 1)$, find $\text{Var}(X)$.

Solution

Since X is random variable which follows an exponential distribution,

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$P(X \leq 1) = P(X > 1)$$

$$1 - P(X > 1) = P(X > 1)$$

$$2P(X > 1) = 1$$

$$P(X > 1) = \frac{1}{2}$$

$$\int_1^{\infty} f(x) dx = \frac{1}{2}$$

$$\int_1^{\infty} \lambda e^{-\lambda x} dx = \frac{1}{2}$$

$$\lambda \left| \frac{e^{-\lambda x}}{-\lambda} \right|_1^{\infty} = \frac{1}{2}$$

$$- \left| e^{-\lambda x} \right|_1^{\infty} = \frac{1}{2}$$

$$-(e^{-\infty} - e^{-\lambda}) = \frac{1}{2}$$

$$e^{-\lambda} = \frac{1}{2}$$

$$\frac{1}{e^{\lambda}} = \frac{1}{2}$$

$$e^{\lambda} = 2$$

$$\lambda = \log_e 2$$

$$\text{Var}(X) = \frac{1}{\lambda^2} = \frac{1}{(\log_e 2)^2}$$

Example 6

The mileage which car owners get with a certain kind of radial tire is a random variable having an exponential distribution with mean 4000 km. Find the probabilities that one of these tires will last (i) at least 2000 km (ii) at most 3000 km.

Solution

Let X be the random variable which denotes the mileage obtained with the tire.

$$\text{Mean } \mu = \frac{1}{\lambda} = 4000 \text{ km}$$

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$= \frac{1}{4000} e^{-\frac{1}{4000}x}, \quad x > 0$$

$$(i) \quad P(X \geq 2000) = \int_{2000}^{\infty} f(x) dx$$

$$= \int_{2000}^{\infty} \frac{1}{4000} e^{-\frac{1}{4000}x} dx$$

$$= \frac{1}{4000} \left| \frac{e^{-\frac{1}{4000}x}}{-\frac{1}{4000}} \right|_{2000}^{\infty}$$

$$\begin{aligned}
 &= - \left| e^{-\frac{1}{4000}x} \right|_{2000}^{\infty} \\
 &= -(e^{-\infty} - e^{-0.5}) \\
 &= e^{-0.5} \\
 &= 0.6065
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad P(X \leq 3000) &= \int_0^{3000} f(x) dx \\
 &= \int_0^{3000} \frac{1}{4000} e^{-\frac{1}{4000}x} dx \\
 &= \frac{1}{4000} \left| \frac{e^{-\frac{1}{4000}x}}{-\frac{1}{4000}} \right|_0^{3000} \\
 &= - \left| e^{-\frac{1}{4000}x} \right|_0^{3000} \\
 &= -(e^{-0.75} - e^0) \\
 &= -e^{-0.75} + 1 \\
 &= 0.5270
 \end{aligned}$$

Example 7

If the number of kilometers that a car can run before its battery wears out is exponentially distributed with an average value of 10000 km and if the owner desires to take a 5000 km trip, what is the probability that he will be able to complete his trip without having to replace the car battery. Assume that the car has been used for same time.

Solution

Let X be the random variable which denotes the number of kilometers that a car can run before its battery wears out.

$$\begin{aligned}
 \text{Mean } \mu &= \frac{1}{\lambda} = 10000 \\
 f(x) &= \lambda e^{-\lambda x}, \quad x > 0 \\
 &= \frac{1}{10000} e^{-\frac{1}{10000}x}, \quad x > 0
 \end{aligned}$$

$$\begin{aligned}
P(X > 5000) &= \int_{5000}^{\infty} f(x) dx \\
&= \int_{5000}^{\infty} \frac{1}{10000} e^{-\frac{1}{10000}x} dx \\
&= \frac{1}{10000} \left| \frac{e^{-\frac{1}{10000}x}}{-\frac{1}{10000}} \right|_{5000}^{\infty} \\
&= - \left| e^{-\frac{1}{10000}x} \right|_{5000}^{\infty} \\
&= -(e^{-\infty} - e^{-0.5}) \\
&= e^{-0.5} \\
&= 0.6065
\end{aligned}$$

Example 8

The average time it takes to serve a customer at a petrol pump is 6 minutes. The service time follows exponential distribution. Calculate the probability that

- (i) A customer will take less than 2 minutes to complete the service.
- (ii) A customer will take between 4 and 5 minutes to get the service.
- (iii) A customer will take more than 10 minutes for his service.

Solution

Let X be the random variable which denotes the service time.

$$\begin{aligned}
\text{Mean } \mu &= \frac{1}{\lambda} = 6 \\
f(x) &= \lambda e^{-\lambda x}, x > 0 \\
&= \frac{1}{6} e^{-\frac{1}{6}x}, x > 0
\end{aligned}$$

$$\begin{aligned}
\text{(i) } P(X < 2) &= \int_0^2 f(x) dx \\
&= \int_0^2 \frac{1}{6} e^{-\frac{1}{6}x} dx
\end{aligned}$$

$$\begin{aligned} &= \frac{1}{6} \left| \frac{e^{-\frac{1}{6}x}}{-\frac{1}{6}} \right|_0^2 \\ &= - \left| e^{-\frac{1}{6}x} \right|_0^2 \\ &= -(e^{-\frac{1}{3}} - e^0) \\ &= -e^{-\frac{1}{3}} + 1 \\ &= 0.2835 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad P(4 < X < 5) &= \int_4^5 f(x) dx \\ &= \int_4^5 \frac{1}{6} e^{-\frac{1}{6}x} dx \\ &= \frac{1}{6} \left| \frac{e^{-\frac{1}{6}x}}{-\frac{1}{6}} \right|_4^5 \\ &= - \left| e^{-\frac{1}{6}x} \right|_4^5 \\ &= - \left(e^{-\frac{5}{6}} - e^{-\frac{2}{3}} \right) \\ &= 0.0788 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad P(X > 10) &= \int_{10}^{\infty} f(x) dx \\ &= \int_{10}^{\infty} \frac{1}{6} e^{-\frac{1}{6}x} dx \\ &= \frac{1}{6} \left| \frac{e^{-\frac{1}{6}x}}{-\frac{1}{6}} \right|_{10}^{\infty} \\ &= - \left| e^{-\frac{1}{6}x} \right|_{10}^{\infty} \end{aligned}$$

$$\begin{aligned}
 &= -\left(e^{-\infty} - e^{-\frac{10}{6}}\right) \\
 &= e^{-\frac{10}{6}} \\
 &= 0.1889
 \end{aligned}$$

Example 9

The length of time X to complete a job is exponentially distributed with

$E(X) = \mu = \frac{1}{\lambda} = 10$ hours. (i) Compute the probability of job completion between two consecutive jobs exceeding 20 hours. (ii) The cost of job completion is given by $C = 4 + 2X + 2X^2$. Find the expected value of C .

Solution

Let X be a random variable which denotes the length of time to complete a job.

$$\begin{aligned}
 E(X) = \mu &= \frac{1}{\lambda} = 10 \\
 f(x) &= \lambda e^{-\lambda x} \\
 &= \frac{1}{10} e^{-\frac{1}{10}x}
 \end{aligned}$$

$$\begin{aligned}
 \text{(i) } P(X > 20) &= \int_{20}^{\infty} f(x) dx \\
 &= \int_{20}^{\infty} \frac{1}{10} e^{-\frac{1}{10}x} dx \\
 &= \frac{1}{10} \left[\frac{e^{-\frac{1}{10}x}}{-\frac{1}{10}} \right]_{20}^{\infty} \\
 &= - \left[e^{-\frac{1}{10}x} \right]_{20}^{\infty} \\
 &= -(e^{-\infty} - e^{-2}) \\
 &= e^{-2} \\
 &= 0.1353
 \end{aligned}$$

(ii) For an exponential random variable,

$$E(X) = \mu = \frac{1}{\lambda} = 10$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

$$\text{Var}(X) = E(X^2) - \mu^2$$

$$E(X^2) = \text{Var}(X) + \mu^2$$

$$= \frac{1}{\lambda^2} + \frac{1}{\lambda^2}$$

$$= \frac{2}{\lambda^2}$$

$$= 200$$

$$\begin{aligned} E(C) &= E(4 + 2X + 2X^2) \\ &= E(4) + 2E(X) + 2E(X^2) \\ &= 4 + 2(10) + 2(200) \\ &= 424 \end{aligned}$$

Example 10

The time (in hours) required to repair a machine is exponentially distributed with parameter $\lambda = \frac{1}{2}$.

- (i) What is the probability that the repair time exceeds 2 hours?
- (ii) What is the conditional probability that a repair takes at least 11 hours given that its direction exceeds 8 hours?

Solution

Let X be the random variable which denotes the time to repair the machine.

$$\lambda = \frac{1}{2}$$

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$= \frac{1}{2} e^{-\frac{1}{2}x}, \quad x > 0$$

$$\begin{aligned}
 \text{(i)} \quad P(X > 2) &= \int_2^{\infty} f(x) dx \\
 &= \int_2^{\infty} \frac{1}{2} e^{-\frac{1}{2}x} dx \\
 &= \frac{1}{2} \left| \frac{e^{-\frac{1}{2}x}}{-\frac{1}{2}} \right|_2^{\infty} \\
 &= - \left| e^{-\frac{1}{2}x} \right|_2^{\infty} \\
 &= -(e^{-\infty} - e^{-1}) \\
 &= e^{-1} \\
 &= 0.3679
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad P(X \geq 11/X > 9) &= P(X > 3) \quad (\text{By the memoryless property}) \\
 &= \int_3^{\infty} f(x) dx \\
 &= \int_3^{\infty} \frac{1}{2} e^{-\frac{1}{2}x} dx \\
 &= \frac{1}{2} \left| \frac{e^{-\frac{1}{2}x}}{-\frac{1}{2}} \right|_3^{\infty} \\
 &= - \left| e^{-\frac{1}{2}x} \right|_3^{\infty} \\
 &= -(e^{-\infty} - e^{-1.5}) \\
 &= e^{-1.5} \\
 &= 0.2231
 \end{aligned}$$